

Insurance Reserve Platform

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1 Introduction

The objective of this project is to develop an intelligent insurance reserve estimation platform using Physics-Informed Neural Networks (PINNs). The model aims to learn reserve trajectories governed by Thiele's differential equation while incorporating actuarial knowledge into the learning process.

The project extends beyond reserve prediction by including modules for sensitivity analysis, stress testing, and optimization, enabling the trained model to support actuarial decision-making and product analysis.

2 Background

This section provides the theoretical concepts required for the project.

2.1 Life Insurance Reserves

Insurance reserves represent the amount of money an insurer must hold to meet future policy obligations. Accurate reserve estimation is essential for pricing, solvency, and regulatory compliance.

2.2 Thiele's Differential Equation

The reserve dynamics are governed by Thiele's differential equation, which models the evolution of reserves using premiums, mortality, and interest rates.

$$\frac{dV}{dt} = rV + P - \mu(S - V)$$

This equation serves as the governing physics constraint throughout the project.

2.3 Physics-Informed Neural Networks

PINNs combine supervised learning with physical laws by minimizing both prediction error and the residual of the governing differential equation. This allows

the model to produce physically consistent reserve estimates while learning from data.

2.4 Project Objectives

The main objectives of the project are:

- Develop a PINN-based reserve prediction model.
- Validate predictions against the classical actuarial solver.
- Perform sensitivity and elasticity analysis.
- Build optimization workflows for actuarial decision support.

3 Project Familiarization

The initial phase focused on understanding the existing codebase and actuarial concepts before making any modifications.

3.1 Repository Understanding

The repository structure, training pipeline, data flow, and model architecture were studied to understand how different components interacted.

3.2 Insurance Reserve Pipeline

The reserve generation process was analysed, including policy simulation, mortality modelling, classical reserve computation, dataset preparation, model training, and evaluation.

3.3 Codebase Analysis

Individual modules such as the policy simulator, Thiele solver, dataset builder, training loop, evaluation pipeline, and visualization utilities were explored to identify potential areas for improvement.

4 Dataset Generation and Validation

Before training the model, the synthetic dataset generation process was analysed to ensure that the generated policies were suitable for learning.

4.1 Synthetic Policy Generation

Policies were generated using configurable ranges for age, policy term, premium, interest rate, mortality, and sum assured. Reserve trajectories for each policy were obtained using the classical Thiele solver.

4.2 Dataset Validation

Several exploratory data analysis (EDA) plots were generated to study the distributions of policy attributes and verify that the generated dataset covered the intended actuarial space.

4.3 Observations

The analysis revealed inconsistencies in the relationships between premiums, sum assured, and other policy variables. These observations motivated subsequent improvements to feature representation and normalization. Several modifications were made to improve the numerical representation of both the input features and reserve targets.

5 Developments

This section summarizes the major experiments performed during the development of the Insurance Reserve Intelligence Platform. Each experiment is documented with the observed problem, the modification made, and the resulting outcome.

5.1 Runs 1–3 : Initial Training and Stabilization

- **Problem:** The model predicted values close to zero, whereas the target reserves were in the range of tens of thousands, resulting in extremely large data loss.

Modification:

- Applied z-score normalization.
- Changed the training target from raw reserve to the standardized reserve ratio (V/S).
- Normalized all input features.

Outcome: Data loss reduced significantly and stable training became possible.

- **Problem:** Different loss terms contributed unevenly during training.

Modification:

- Tuned the weights of Data Loss.
- Tuned PDE Loss.
- Tuned Boundary Loss.
- Tuned Monotonicity Losses.
- Tuned Smoothness Loss.

Outcome: Training became more balanced with improved convergence.

- **Problem:** Reserve trajectories generated by the PINN did not match the expected actuarial behaviour.

Modification:

- Increased the number of training epochs.
- Updated configuration parameters.
- Added a reserve trajectory comparison plot between the PINN and the classical Thiele solver.

Outcome: Reserve curves became more realistic and model behaviour could be visually monitored.

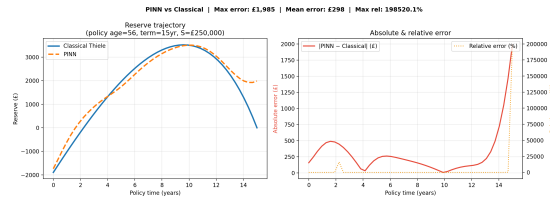


Figure 1: PINN vs Classical Reserve Trajectory

- **Problem:** It was unclear whether the synthetic portfolio resembled realistic insurance data.

Modification:

- Generated exploratory data analysis (EDA) plots.
- Analysed distributions of policy age, term, premium, interest rate and sum assured.

Outcome: Helped verify the realism of the simulated policy portfolio.

5.2 Runs 4–8 : Model Refinement and Validation

- **Problem:** There was no systematic way to evaluate whether the PINN was learning correctly.

Modification:

- Added a validation framework.
- Introduced validation plots for model performance assessment.

Outcome: Enabled quantitative evaluation of model accuracy, physics consistency and generalization.

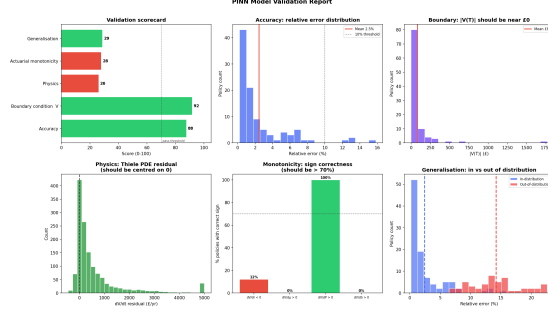


Figure 2: Initial Validation Dashboard

- **Problem:** Data loss dominated training while physics-based constraints had very little influence.

Modification:

- Rebalanced the weights of data, PDE and boundary losses.

Outcome: Improved balance between data fitting and physics-informed learning.

- **Problem:** PDE loss frequently became extremely large or diverged because reserve values were represented in raw currency.

Modification:

- Reformulated the PDE using the normalized reserve ratio (V/S) instead of raw reserve values.

Outcome: PDE residual became numerically stable and training convergence improved.

- **Problem:** Reserve predictions did not consistently increase with increasing sum assured.

Modification:

- Added a Sum Assured Monotonicity Loss.

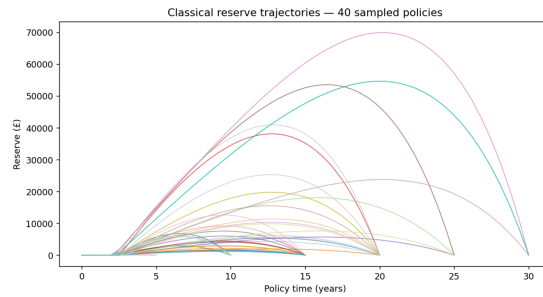
Outcome: Encouraged reserves to increase with policy coverage, improving actuarial consistency.

- **Problem:** Existing EDA plots only described policy distributions and did not explain reserve behaviour.

Modification:

- Added reserve dependency plots for:
 - * Age
 - * Mortality
 - * Premium
 - * Interest Rate
 - * Sum Assured

Outcome: Improved understanding of how reserves vary with different policy characteristics.



- **Problem:** Training was unstable and validation performance fluctuated significantly.

Modification:

- Tuned the learning rate.
- Updated the learning rate scheduler.
- Adjusted gradient clipping.
- Increased dataset size.

Outcome: Training became more stable with improved convergence.

- **Problem:** Autograd gradients were extremely small, making monotonicity constraints ineffective.

Modification:

- Replaced autograd-based gradients with finite-difference perturbations for:
 - * Mortality

- * Interest Rate
- * Sum Assured

Outcome: Produced stronger and more reliable monotonicity signals during validation.

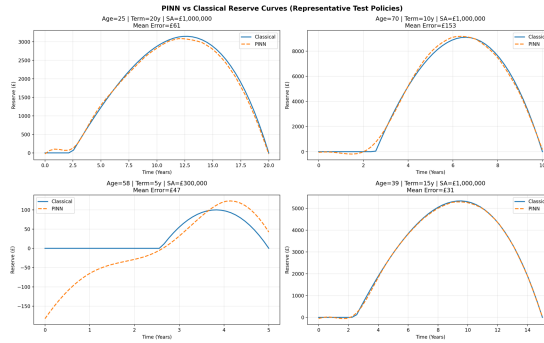
5.3 Runs 9-15 : Sensitivity Analysis

- **Problem:** Negative reserve values caused issues in optimization and capital calculation modules.

Modification:

- Applied a minimum reserve value of zero before returning solver outputs.

Outcome: Improved downstream numerical stability.



- **Problem:** Model evaluation relied on limited metrics.

Modification:

- Added:
 - * Mean Squared Error (MSE)
 - * Mean Absolute Error (MAE)
 - * Root Mean Squared Error (RMSE)
 - * R^2 Score

Outcome: Enabled more comprehensive evaluation of prediction quality.

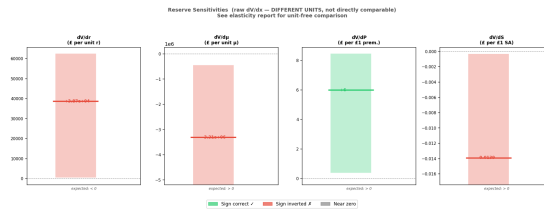
- **Problem:** Model explainability was limited.

Modification:

- Implemented sensitivity analysis by computing reserve derivatives with respect to:

- * Interest Rate
- * Mortality
- * Premium
- * Sum Assured

Outcome: Generated the first sensitivity reports, although several derivative signs were inconsistent with actuarial expectations.



- **Problem:** Raw sensitivities had different units and could not be directly compared.

Modification:

- Added elasticity calculations to convert sensitivities into unit-free percentage changes.

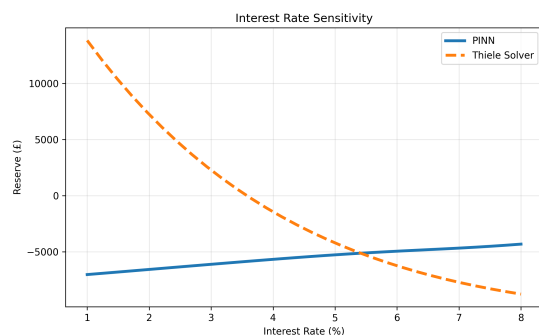
Outcome: Enabled direct comparison of feature importance across variables.

- **Problem:** It was unclear whether incorrect sensitivities originated from the PINN or from the classical solver.

Modification:

- Compared PINN derivatives with derivatives obtained from the classical Thiele solver.

Outcome: Enabled direct benchmarking of learned sensitivities against the actuarial solution.

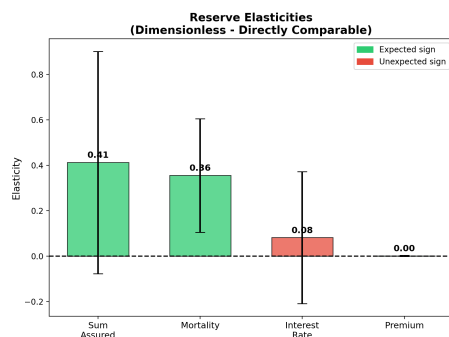


- **Problem:** Premium appeared to dominate the learning process, making it difficult to identify the true source of sensitivity errors.

Modification:

- Retrained the model with premium fixed to zero.
- Repeated sensitivity analysis while keeping all other variables unchanged.

Outcome: Confirmed that premium was a dominant learning signal and that the remaining sensitivity errors were related to model learning rather than evaluation bugs.



- **Problem:** Raw premium remained strongly coupled with Sum Assured, preventing independent sensitivity analysis.

Modification:

- Replaced Premium with the normalized ratio (P/S) across:
 - * Dataset
 - * Model
 - * Loss functions

- * Evaluator
- * Optimization modules

Outcome: Allowed premium and sum assured to be treated as independent variables during sensitivity analysis.

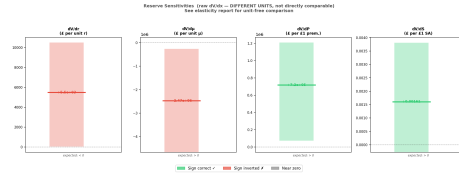


Figure 3: Current Sensitivity

- **Problem:** Changing one feature unintentionally altered other dependent policy variables.

Modification:

- Constructed a fixed reference policy.
- Varied only one feature at a time during sensitivity evaluation.

Outcome: Produced true partial derivatives, although the overall sensitivity trends remained largely unchanged.

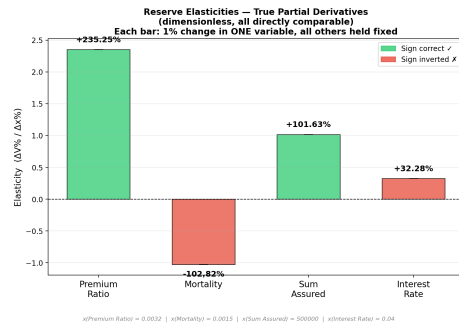


Figure 4: Current Elasticity

- **Current Status**

Current sensitivity results indicate:

- Premium Sensitivity Correct
- Sum Assured Sensitivity Correct

- Mortality Sensitivity × Incorrect
- Interest Rate Sensitivity × Incorrect

The remaining work focuses on improving the physics learning so that all sensitivities follow actuarial expectations.